Closed-Loop Transfer Function and State Diagrams

Embedded Control of Robotics – worked examples

1. Unity feedback loop

A block G(s) with its output C(s) fed back with gain 1 to a summing junction at the input R(s).

Negative feedback

1. Error at the summing junction: $E = R - C$
2. Block equation: $C = G \cdot E$
3. Substitute: $C = G(R - C) = GR - GC$
4. Collect C terms: $C + GC = GR \rightarrow C(1 + G) = GR$
5. Divide:

```
C(s)/R(s) = G / (1 + G)
```

Positive feedback

```
E = R + C   →   C = G(R + C)   →   C(1 − G) = GR
C(s)/R(s) = G / (1 − G)
```

The sign written at the summing junction for the feedback signal shows which case applies. A minus sign means negative feedback, which is the usual case.

2. Transfer function from a state diagram with two integrators

Each 1/s block is an integrator. Name the integrator outputs as the states: x_1 is the output of the last integrator and x_2 is its input (so $x_2 = \dot{x}_1$).

```
ẋ₁ = x₂
ẋ₂ = u − 4x₂ − 3x₁
y  = 2x₁ + 2x₂
```

Laplace transform (zero initial conditions)

```
sX₁ = X₂
sX₂ = U − 4X₂ − 3X₁
```

Substitute $X_2 = sX_1$ into the second equation:

```
s²X₁ = U − 4sX₁ − 3X₁
(s² + 4s + 3) X₁ = U
X₁/U = 1 / (s² + 4s + 3)        X₂/U = s / (s² + 4s + 3)
```

Output

```
Y/U = 2·X₁/U + 2·X₂/U = (2 + 2s) / (s² + 4s + 3)
s² + 4s + 3 = (s + 1)(s + 3)
Y/U = 2(s + 1) / ((s + 1)(s + 3))
```

```
G(s) = Y(s)/U(s) = 2 / (s + 3)
```

The zero at -1 cancels the pole at -1, so the system behaves like a first-order system with a pole at -3 and a DC gain of 2/3.